Technical Proposal

Project Name: Customer Churn Prediction & Automated Retraining System

Team Name: 6 Guys

Team Leader: Abdullah Mohame

Team Members: Farouk Mohamed, Youssif Hamada, Mahmoud Khaled, Sayed Hany

1. Executive Summary

This proposal outlines the development of a production-grade customer churn prediction system. The system will ingest streaming customer behavior data, build and serve a churn prediction model, and implement automated retraining based on data drift. Our goal is to transition from reactive customer retention to proactive, data-driven forecasting, providing stakeholders with actionable dashboards and alerts to reduce churn.

2. Problem Statement and Objectives

Customer churn directly impacts business profitability. Traditional retention strategies often fail to identify at-risk customers before they cancel their service. We propose a machine learning solution to detect complex usage patterns and predict churn probability in real time.

The primary objectives of this project are to:

- Gather and prepare churn data from various sources, including internal CRM, logs, and billing.
- Build and evaluate churn prediction models with a production focus.
- Deploy the churn model as a scalable prediction service on Azure.
- Implement MLOps best practices for lifecycle management and business integration.

3. Proposed Methodology and Scope

The project will follow a structured lifecycle, advancing from raw data to a deployed API with an automated monitoring loop.

Phase 1: Data Preparation and Exploration

The team will collect historical and streaming customer transactional and usage data from SQL databases, logs, or Azure Blob Storage. We will clean the features, handle null values, and address class imbalance using techniques such as SMOTE or weighting. Exploratory Data Analysis (EDA) will visualize churn correlates, such as tenure and usage patterns, while investigating potential bias.

Phase 2: Model Development and Evaluation

We will implement classification algorithms, specifically evaluating Gradient Boosting, XGBoost, and Random Forest. The evaluation will rely on business-relevant classification metrics, including the confusion matrix, ROC-AUC, and precision-recall curves.

Phase 3: Cloud Deployment

The final model will be deployed via Azure Machine Learning as a REST endpoint using Docker containers for standardized serving. We will build a REST API to receive customer data and return churn risk scores in real time.

Phase 4: MLOps and Dashboard Integration

We will use MLflow or Azure ML for experiment tracking. A web dashboard built with Streamlit will display churn predictions and business KPIs. We will also establish automated retraining pipelines triggered by data drift metrics.

4. Technical Specifications

- **Programming Languages & Libraries:** Python, Pandas, NumPy, Scikit-learn, XGBoost.
- **Modeling Approaches:** Gradient Boosting, XGBoost, Random Forest.
- **Evaluation Metrics:** ROC-AUC, precision-recall curves, confusion matrix.
- **Deployment & MLOps:** Azure ML, Docker, REST API, MLflow, Streamlit.

5. Project Timeline and Deliverables

The estimated 10-week timeline is structured to deliver incremental value and ensure rigorous evaluation.

- **Weeks 1-2:** Data Collection, Preprocessing, and Exploration.
- **Weeks 3-5:** Model Development, Hyperparameter Tuning, and Evaluation.
- **Weeks 6-7:** Cloud Deployment and API Integration.
- **Weeks 8-9:** MLOps Setup, Monitoring, and Web Dashboard.
- **Week 10:** Final Documentation, Future Roadmap, and Presentation.

6. Team Responsibility Matrix

Team Member	Role & Expected Contribution
Abdullah Mohame (Team Leader)	Coordinates team activities, tracks milestones, and ensures integration. Co-develops ML models (Gradient Boosting, XGBoost) and evaluates performance using ROC-AUC and precision-recall curves.
Sayed Hany	**Data Collection & Cleaning:** Collects transactional data. Cleans data using Pandas/NumPy, manages missing values, and handles class imbalance using SMOTE or weighting.
Farouk Mohamed	**EDA & Feature Engineering:** Conducts exploratory analysis to visualize churn correlates (tenure, usage). Investigates bias, performs feature selection, and prepares structured inputs.
Youssif Hamada	**Machine Learning Modeling:** Collaborates on model training (Random Forest, ensemble techniques). Performs hyperparameter tuning and threshold optimization for business impact.
Mahmoud Khaled	**Dashboard & Integration:** Develops an interactive web dashboard using Streamlit for predictions and KPIs. Integrates the interface with the Azure ML REST API.